

Section 11

Theorem 0.1 (Lie Group to Lie Algebra). *This is an adjusted version of Chatterjee (2016).*

There exists a constant r_0 depending only on N and d such that for all $r \leq \frac{r_0}{4}$, we have

$$(2-e^{6r})^{C_N} \int_{b(0,r)} f(e^{iH}) d\lambda(H) \leq \int_{B(0,2r)} f(U) d\sigma(U) \leq (e^{6r})^{C_N} \int_{b(0,3r)} f(e^{iH}) d\lambda(H).$$

where $f : U(N) \rightarrow [0, \infty]$ is a Borel measurable function.

By Theorem 0.1, we have proved the Yang-Mills conjecture.

Chatterjee, Sourav. 2016. “The Leading Term of the Yang-Mills Free Energy.”